CLIMATE CONTROL UNIT [MANUAL AIR CONDITIONER]

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Purpose

- The climate control unit controls the air conditioner in conjunction with the A/C operations performed by the occupants.

Function

- The climate control unit outputs an operation signal for each switch or dial. (See MANUAL AIR CONDITIONER SYSTEM [MANUAL AIR CONDITIONER].)

Dial set position detection function

- The climate control unit detects the set position of each operation dial and outputs the following signals to the instrument cluster.
- The instrument cluster determines each control based on each dial set position signal from the climate control unit.

RECIRCULATE/FRESH lever type

Dial set position	Output signal	Control item
Set position of temperature setting dial (Without heater system)	Set position signal	• A/C compressor control (See MANUAL AIR CONDITIONER SYSTEM [MANUAL AIR CONDITIONER].)
Airflow volume control dial is at 1st or higher	Blower ON signal	• A/C compressor control (See MANUAL AIR CONDITIONER SYSTEM [MANUAL AIR CONDITIONER].)
Airflow volume control dial is at 3rd or 4th	Blower speed signal	• Generator output control (See GENERATOR CONTROL [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) (See GENERATOR CONTROL [SKYACTIV-D 1.5].)

RECIRCULATE/FRESH switch type

Dial set position	Output signal	Control item
MODE dial is at defroster	Defroster position signal	• Air intake control (See MANUAL AIR CONDITIONER SYSTEM [MANUAL AIR CONDITIONER].) • i-stop (engine stop control) permission determination (With i-stop) (See MANUAL AIR CONDITIONER SYSTEM [MANUAL AIR CONDITIONER].)
Temperature setting dial is at MAX COLD (With MAX A/C control)	MAX COLD position signal	• A/C compressor control (See MANUAL AIR CONDITIONER SYSTEM [MANUAL AIR CONDITIONER].)
Airflow volume control dial is at 1st or higher	Blower ON signal	• A/C compressor control (See MANUAL AIR CONDITIONER SYSTEM [MANUAL AIR CONDITIONER].) • i-stop (engine stop control) permission determination (With i-stop) (See MANUAL AIR CONDITIONER SYSTEM [MANUAL AIR CONDITIONER].)
Airflow volume control dial is at 3rd or 4th	Blower speed signal	• Generator output control (See GENERATOR CONTROL [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) (See GENERATOR CONTROL [SKYACTIV-D 1.5].)

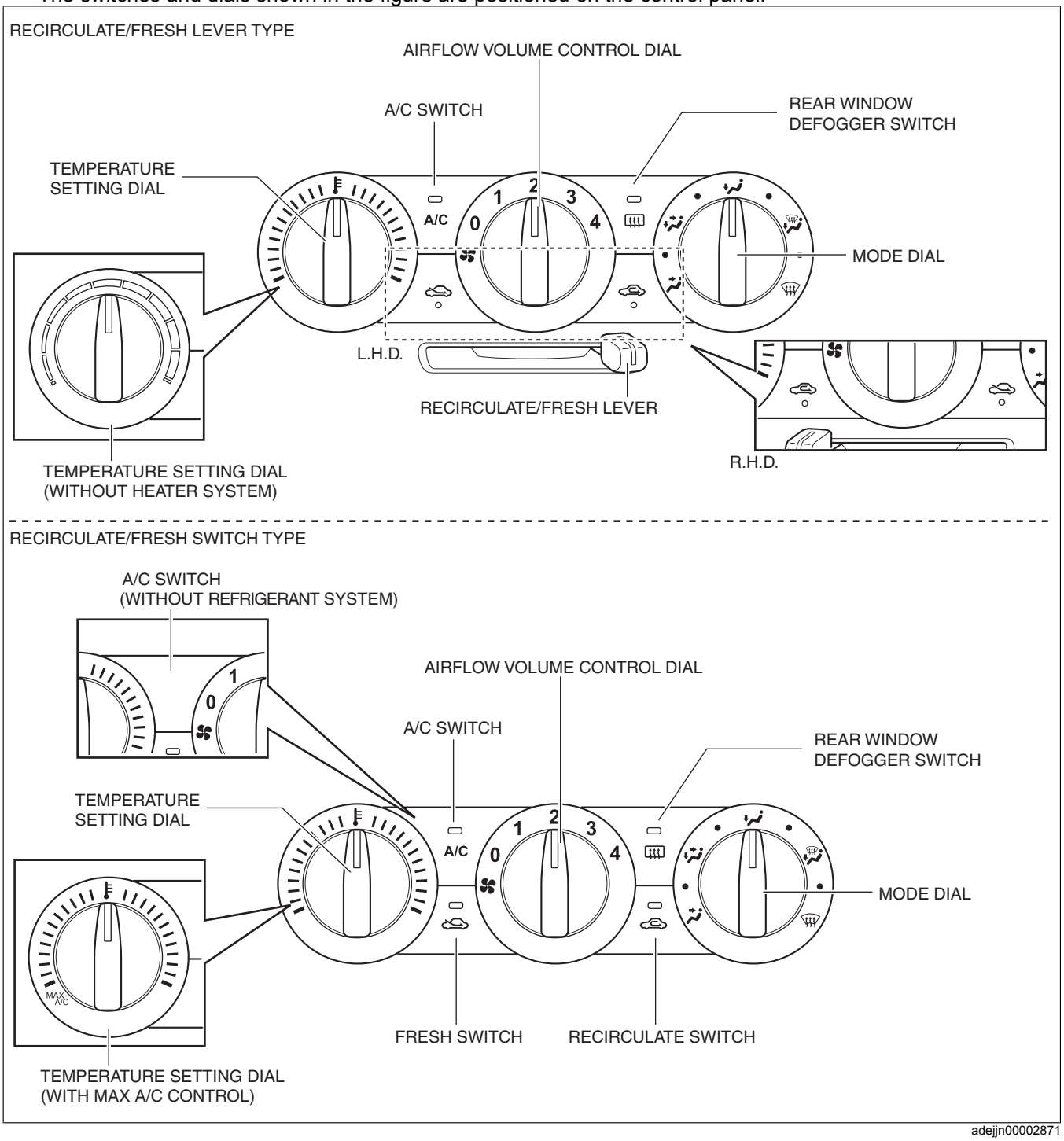
Construction

- The climate control unit has adopted a wire type to the following controls.
  - — Air mix door (temperature setting dial)
  - — Air intake door (RECIRCULATE/FRESH lever (RECIRCULATE/FRESH lever type))
  - — Airflow mode door (MODE dial)
- The climate control unit consists of a control panel and a control unit.
- The MS-CAN terminal resistance is integrated into the climate control unit.

Note

- The terminal resistance integrated into the climate control unit is used by the MS-CAN transmission, but CAN transmission is not used in the manual air conditioner system.

- The switches and dials shown in the figure are positioned on the control panel.



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Operation

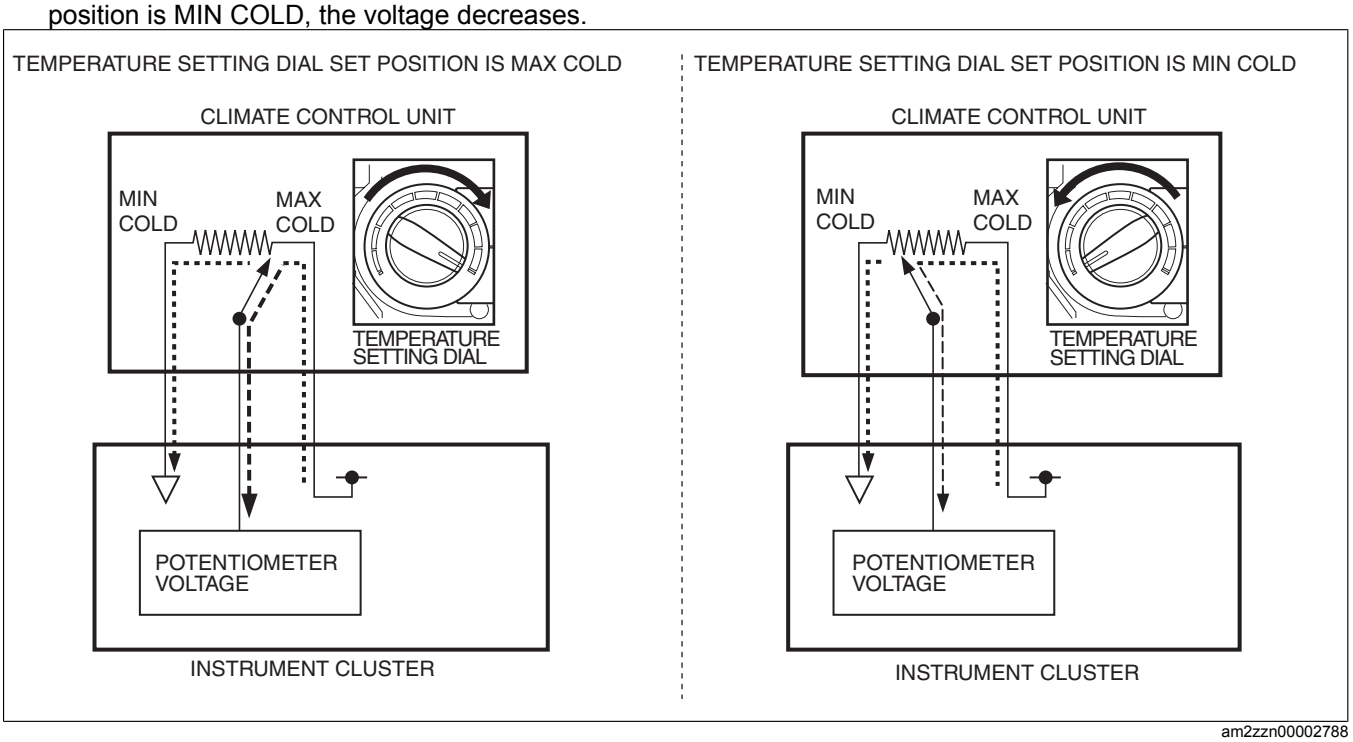
RECIRCULATE/FRESH lever type

- The climate control unit performs the controls shown in the table according to the switch/dial operation.

Operation switches/dials		Airflow temperature control	Airflow volume control	Airflow mode control	A/C compressor control
		Control before switch operation	Control before switch operation	Control before switch operation	Control before switch operation
		Manual control	Manual control	Manual control	Manual control
Airflow volume control dial	OFF	No change	OFF	No change	OFF
	1—4		Manual control		No change
MODE dial	VENT	No change	No change	Manual control	No change
	BI-LEVEL				
	HEAT				
	DEF/HEAT				
	DEFROSTER				
Temperature setting dial	COLD	No change	No change	No change	No change
	HOT				
A/C switch		No change	No change	No change	Manual control
Rear window defogger switch		No change	No change	No change	No change

Temperature setting dial set position detection function (Without heater system)

- For the temperature setting dial set position detection function, the potentiometer integrated with the climate control unit is linked to the temperature setting dial operation.
- When the temperature setting dial set position is MAX COLD, the potentiometer voltage increases. When the position is MIN COLD, the voltage decreases.



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RECIRCULATE/FRESH switch type

- The climate control unit performs the controls shown in the table according to the switch/dial operation.

Operation switches/dials		Airflow temperature control	Airflow volume control	Airflow mode control	
		Control before switch operation	Control before switch operation	Control before switch operation	
		Manual control	Manual control	Manual control	Defroster control
Airflow volume control dial	OFF	No change	OFF	No change	No change
	1—4		Manual control		

Operation switches/dials		Airflow temperature control	Airflow volume control	Airflow mode control	
		Control before switch operation	Control before switch operation	Control before switch operation	
		Manual control	Manual control	Manual control	Defroster control
MODE dial	VENT	No change	No change	Manual control	Manual control
	BI-LEVEL				
	HEAT				
	DEF/HEAT				
	DEFROSTER				
Temperature setting dial	MAX COLD*1	No change	No change	No change	No change
	Except MAX COLD				
A/C switch		No change	No change	No change	No change
RECIRCULATE switch		No change	No change	No change	No change
FRESH switch		No change	No change	No change	No change
Rear window defogger switch		No change	No change	No change	No change

*1 : With MAX A/C control

Operation switches/dials		A/C compressor control		Air intake control		
		Control before switch operation		Control before switch operation		
		Manual control	MAX A/C control*1	Manual control	Defroster control	MAX A/C control*1
Airflow volume control dial	OFF	OFF	OFF	No change	No change	No change
	1—4	No change	No change			
MODE dial	VENT	No change*1, *2	No change	No change*1, *2	Manual control*1, *2	No change
	BI-LEVEL					
	HEAT	No change	Manual control	No change	Manual control	Manual control
	DEF/HEAT					
	DEFROSTER			Defroster control	No change	Defroster control
Temperature setting dial	MAX COLD*1	No change *1, *2	No change	No change*1, *2	No change	No change
	Except MAX COLD	No change	Manual control	No change		Manual control
A/C switch		Manual control	Manual control	No change	No change	No change
RECIRCULATE switch		No change	No change	Manual control	No change	Manual control
FRESH switch		No change	No change	Manual control	No change	Manual control
Rear window defogger switch		No change	No change	No change	No change	No change

*1 : With MAX A/C control

*2 : When the MAX A/C control conditions are met, it switches to MAX A/C control.